

Functional Coding in Web



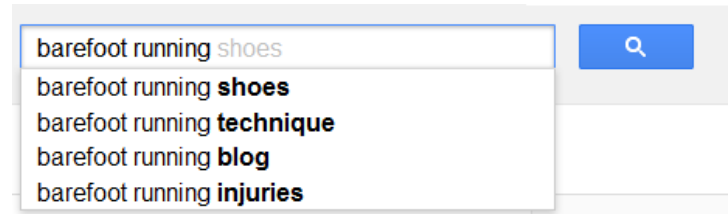
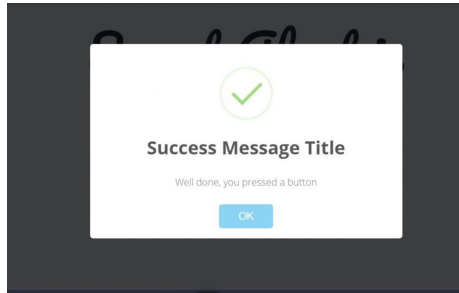
Online surveys are helpful and convenient to collect data in a well-structured format. They have a wider reach as compared to traditional methods of collecting data.



The internet has become an integrated part of our lives. When we want any kind of information, we automatically pick up the closest device and browse the internet to look for it. Millions of websites exist online, each with its own visual aspects and functionalities.

Some common functionalities of a website are:

- Suggestions appear while we type in a search box.
- A dialog box is displayed when we click a button.
- Web contents appear along with animations and effects.
- HTML forms display an error message when we enter incorrect or incomplete information.



Web developers often add these functionalities with JavaScript.

JavaScript



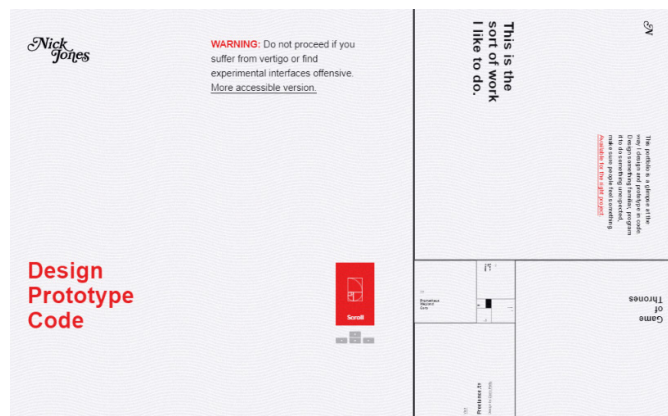
JavaScript is a programming language used to make interactive and dynamic websites. JavaScript enables users to interact with a webpage, mainly through HTML Forms. Hundreds of dynamic functionalities can be implemented using JavaScript, including browsing data and receiving search results. JavaScript is also referred to as a client-side scripting language because programs written in this language are executed on the browser itself (just like HTML and CSS) and not on a remote server.

Advantages of using JavaScript

Like other languages, JavaScript has its own perks.

- Apart from building interactive websites, JavaScript is also used to create web applications, games, mobile apps, and desktop applications.
- It is used by over 94% of all the internet's websites.
- JavaScript will remain relevant until users continue to interact with the web.
- JavaScript's biggest advantage is the ability to support all modern browsers and produce comparable results.
- It is easy to learn, with powerful frameworks.

Examples of JavaScript-based websites



Bytes

JavaScript was initially created as a scripting language for Netscape by Brendan Eich in 1995, when it was known as LiveScript. Netscape later renamed it to JavaScript to become an acceptable companion to the Java language.

Activity

Implement some basic dynamic (interactive) functionalities using JavaScript in a website. The website should feature a simple design and a dynamic counter functionality to keep track of how many users have signed the “Save Ocean” petition.

Through this activity, we will also understand how to use HTML forms to collect user input.

1 Create files using VS Code

- a. Open VS Code and create a new file **index.html**. Save it in your desired location.

```
1 <html>
2   <head>
3     <title> Save the Ocean </title>
4   </head>
5
6   <body>
7
8   </body>
9 </html>
```

- b. Create a div container with a class “**main**”.
This will behave as the main container, inside which all our other contents will be placed.

```
<html>
  <head>
    <title> Save the Ocean </title>
  </head>
  <body>
    <div class="main">
    </div>
  </body>
</html>
```

2 Create an external CSS file for CSS codes

- a. Create a **style.css** file in the same location and then link it with the **index.html** file using the **link tag**.

```
<head>
  <title> Save the Ocean </title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
```

Activity

Recall:

CSS is a website styling language . It provides properties to design the page and arrange its contents according to a layout.

CSS properties are set in the following format (syntax):

`<property_name> : <value>`

3 Create and design the website

For the main container, we will set basic properties like a default font, container width, and centre alignment of the content.

<pre><body> <div class="main"> </div> </body></pre>	<pre>2 3 .main{ 4 font-family:Arial; 5 width: 85%; 6 text-align: center; 7 margin: auto; 8 } 9 10 11</pre>
---	--

Create a header section

- Create another div container inside the main container with a class “**header**”. This section will display a title along with a background image.

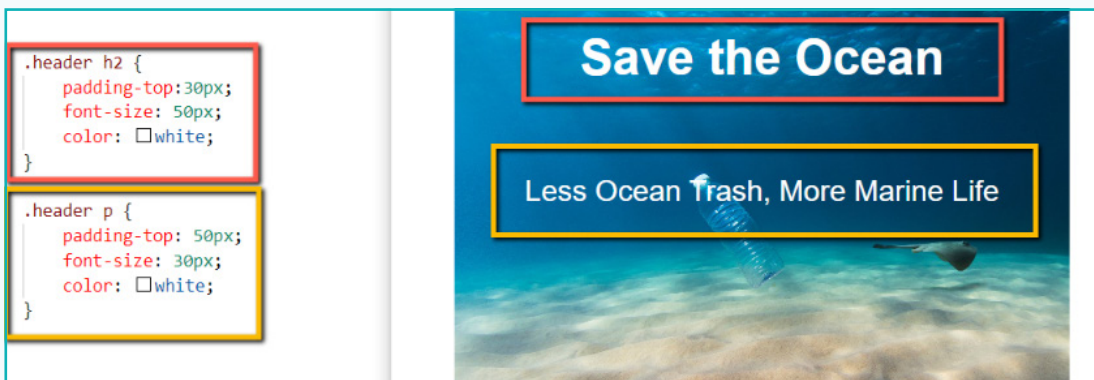
<pre><div class="header"> <h2> Save the Ocean </h2> <p> Less Ocean Trash, More Marine Life </p> </div></pre>	Save the Ocean Less Ocean Trash, More Marine Life
--	---

Activity

- b. In the **style.css** file, use CSS to set a background image for the header section with a fixed height. Use the **background-size** and **background-position** property to align the image as you see fit.

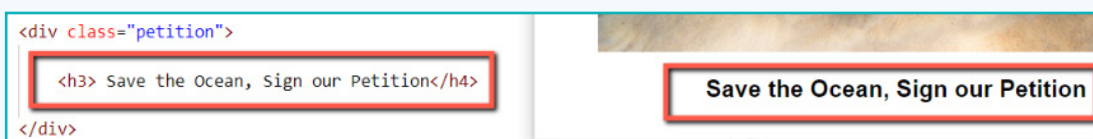


- c. Next, arrange the text elements within the **header** container. To access them in the **style.css** file, specify the name of the class (**header**) and then the heading or paragraph tag. **.header h2** and **.header p** will specify the header and paragraph tag inside the header container. Any other **<h2>** or **<p>** outside the header container will not be affected.



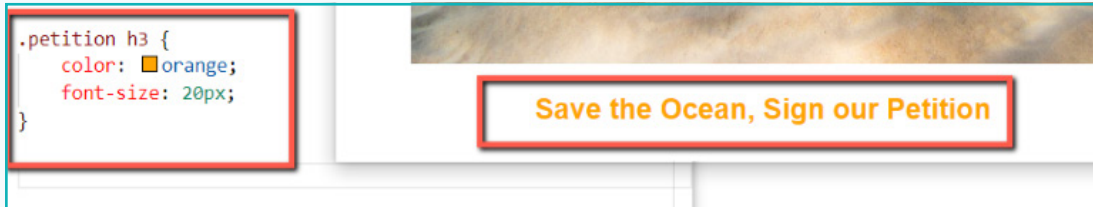
Create the petition section

- d. Create another container with the class **"petition"** and add content inside it.



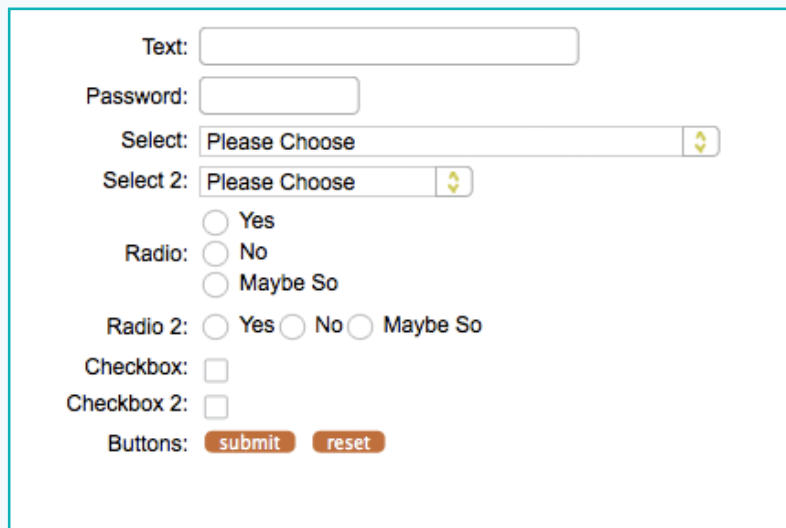
Activity

- e. In the style.css file, change the colour of the heading to “orange” and increase the font size a little.



Add forms for user interaction

At this point, we can start to make our site interactive and the simplest way to do this is with HTML Forms. Forms make it convenient for us to collect user input. HTML provides form elements like text boxes, radio buttons, checkboxes, drop-down menus, buttons, etc. which we can make use of.

The image shows a web form with the following elements: a "Text:" label followed by a text input field; a "Password:" label followed by a password input field; a "Select:" label followed by a dropdown menu showing "Please Choose"; a "Select 2:" label followed by a dropdown menu showing "Please Choose"; a "Radio:" label followed by three radio buttons labeled "Yes", "No", and "Maybe So"; a "Radio 2:" label followed by three radio buttons labeled "Yes", "No", and "Maybe So"; a "Checkbox:" label followed by a checkbox; a "Checkbox 2:" label followed by a checkbox; and a "Buttons:" label followed by two buttons labeled "submit" and "reset".

To create forms, we use the `<form>` tag in HTML. All form elements are placed inside this tag.

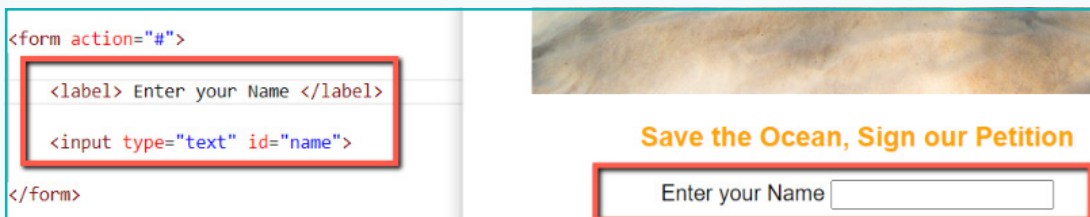
- f. Below the `<h3>` tag, within the “petition” class container, create a **form** tag.

```
<div class="petition">  
  <h3> Save the Ocean, Sign our Petition</h3>  
  <form>  
  </form>  
</div>
```

Activity

- g. Create a textbox for a user to enter their name.

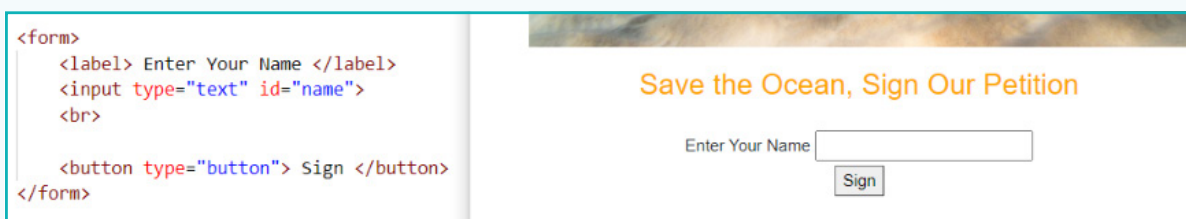
Text boxes are created using the **<input>** tag, with the value of the **type** attribute set to **text**. Create a label using the **<label>** tag so that the user knows what to enter in the text box.



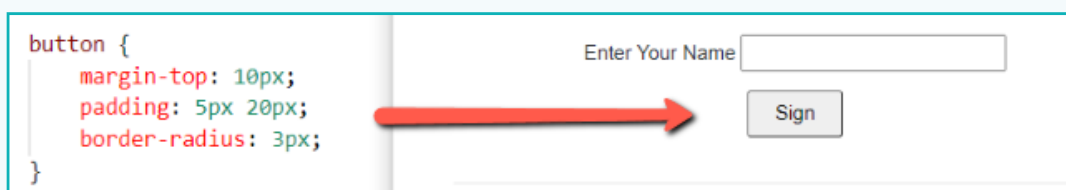
Note:

Notice the attribute id with the value name in the textbox. This will be explained later when we implement JavaScript.

- h. We also need a button to process input data. Create a button using the **<button>** tag. Set the **type** attribute on the button tag to **"button"**.



- i. Additionally, apply CSS properties to align the button as well as make it a bit wider.



Activity

Create the petition counter section

This section will display the total number of users who have signed the petition.

- j. Create a container with a class “**petition-counter**”. Inside the container create a heading using the **<h3>** tag.

```
<div class="petition_counter">
  <h3> Number of people who signed this Petition</h3>
</div>
```

Number of people who signed this
Petition

- k. Create another heading tag **<h2>**. This heading will initially contain 0. Later, we will update this value as and when the user clicks on the **Sign** button.

```
<div class="petition-counter">
  <h3> Number of people who Signed this Petition</h3>
  <h2> 0 </h2>
</div>
```

Number of people who Signed this Petition

0

- l. Add CSS to the counter to make it look attractive.

```
.petition-counter h2{
  font-size: 50px;
  color: grey;
  border-bottom: 5px solid orange;
  width: 100px;
  margin: auto;
}
```

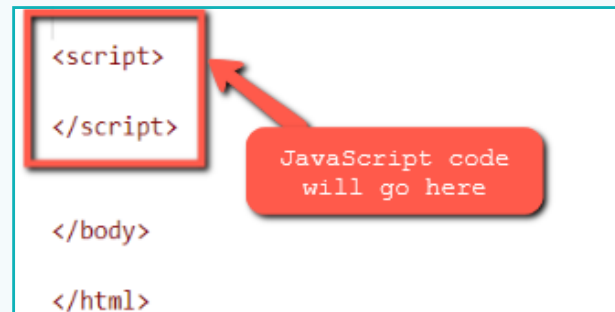
Number of people who Signed this Petition

0

Activity

4 Implement dynamic functionality on the website using JavaScript

To implement dynamic functionalities, we need to integrate JavaScript on a webpage. We use the **<script>** tag to use JavaScript in our application. This tag is usually placed at the bottom of the page, just before the closing **<body>** tag.



Understand variables

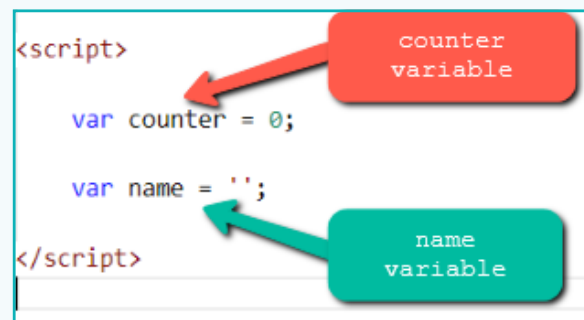
First, we need to store users' names and the total number of petitions signed. We use **variables** to do this.

Variables are containers that store data. To declare a variable in JavaScript, we use the **var** keyword. To assign value to a variable, we use the assignment operator (`=`).

Syntax –

var <variable_name> = value;

- a. Inside the `<script>` tag, create 2 variables `counter` and `name`—using the `var` keyword. Initialize the counter variable to 0 and the name variable as blank (by using single-quotes). Each variable declaration should end with a semi colon.



- b. We need to write JavaScript code to obtain the users' names from the textbox and keep track of how many have signed the petition. To bundle all those codes, we use the function concept.

Activity

Functions in JavaScript

A function is a block of code that performs a specific task. The task may be a simple one like the addition of 2 numbers, or a highly complex one, such as fetching data from a remote server and displaying it on a webpage. Functions also provide a way to bundle code so that it can be reused.

A function is created by using the keyword **function** and then adding a name to it. Code in a function is written between curly brackets.

Syntax – function function_name {
JavaScript Code
}

- c. Define a function called “countPetitions”. To test it out, write an alert(“It Works”) statement inside it.

Note:

alert() is a pre-defined JavaScript function that displays a pop-up box when executed.

```
<script>

    var counter = 0;

    var name = '';

    function countPetitions() {
        alert(" It Works ");
    }

</script>
```

- d. If we view the page on a browser right now, we won't see any pop-up boxes even though we have defined a function for it. This is because we haven't executed the function yet.

Let's associate the **countPetitions()** function with the **Sign** button. This will execute the function whenever the button is clicked.

- e. In the button tag, using the **onclick() handler**, associate the function with the button. We will now be able to execute the **countPetitions()** function when we click the **Sign** button.

```
<div class="petition">
    <h3>Save the Ocean, Sign Our Petition </h3>

    <form action="#">
        <label> Enter Your Name </label>
        <input type="text" id="name">
        <br>

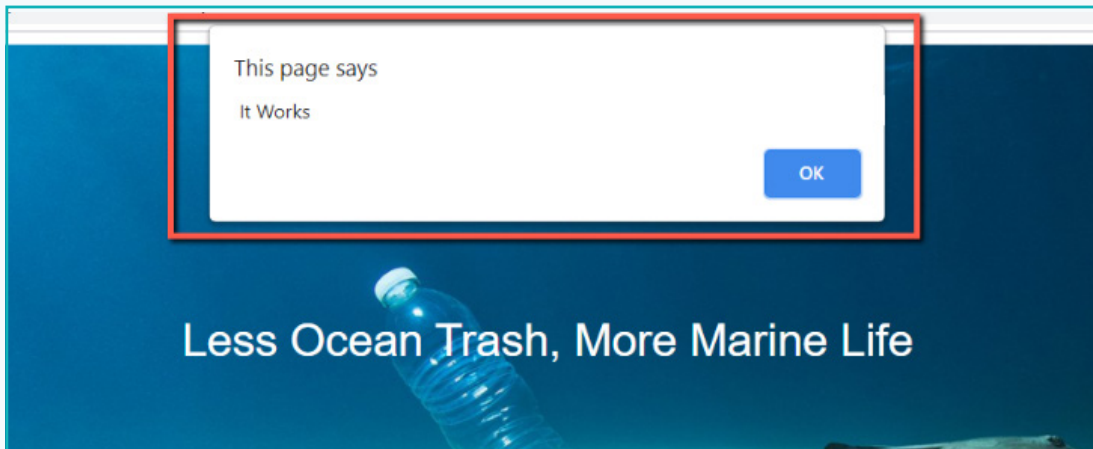
        <button type="button" onclick="countPetitions()"> Sign </button>
    </form>
</div>
```

Function to execute on clicking the button

Event handler

Activity

- f. Click on the button and a pop-up with the text “It Works” will display on the browser.



Now that we have confirmed that our button and its associated function is working, we can remove the **alert()**, and write the actual JavaScript code.

5 Get data from the textbox

To get data from the textbox, first we need to access the textbox. JavaScript provides various pre-defined functions that can be used to access any HTML element present in a webpage. **getElementById()**, **getElementsbyClassName()**, **getElementsbyTagName()**, **getElementsByName()** are some of them. For this website, we will focus on **getElementById()**.

This method, “**getElementById()**” is used to get an element with a specific ID.

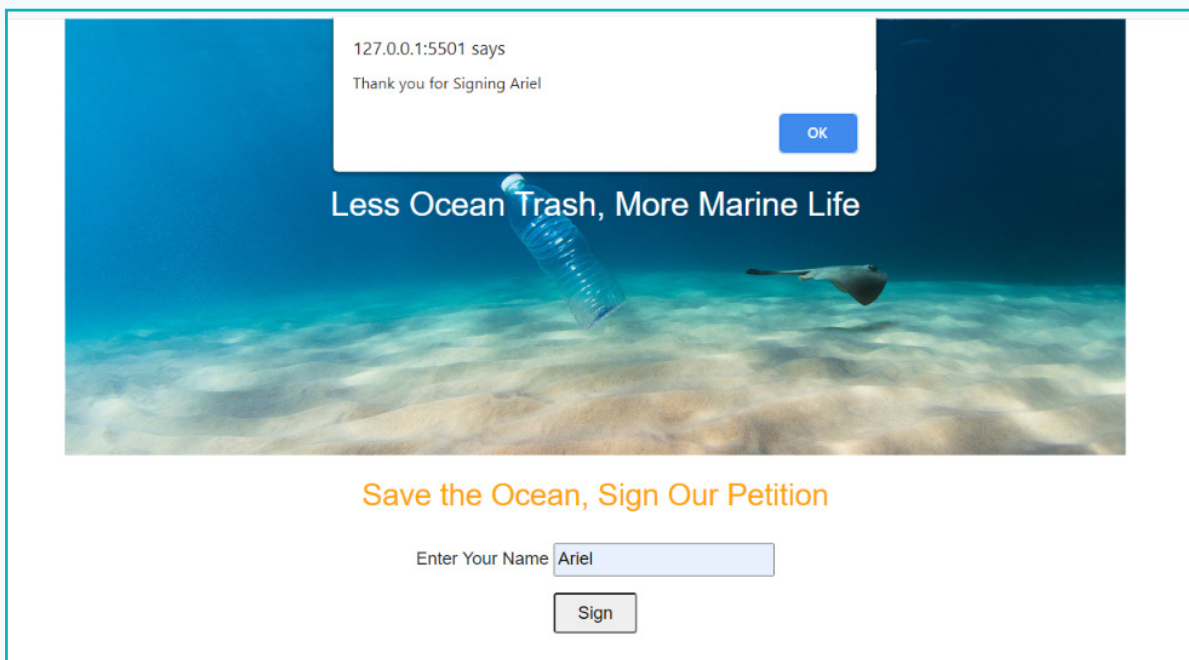
- a. We have already provided an ID value to the textbox. Use that to access the textbox and retrieve the user-entered value from inside it.

```
function countPetitions() {  
    name = document.getElementById('name').value;  
    alert("Thank You for Signing "+name);  
}
```

We used the **alert()** function to display a message for the user which confirms his signature on the petition.

Activity

Enter your own name in the textbox to test it. Run the file and click on the Sign button. The alert box will display a message for you!



6 Maintain a count of users signing the petition

We have already created a variable **counter** to maintain the number of users who have signed up. We need to increase the value of the counter variable whenever a user clicks on the **Sign** button.

- a. Below the **alert()** statement, increment the counter variable by adding 1 to it.

```
function countPetitions() {  
    name = document.getElementById('name').value;  
    alert("Thank You for Signing "+name);  
    counter = counter + 1;  
}
```

Activity

Here, we are simply adding 1 to the existing value of the **counter** variable and then updating the counter variable with the new value.

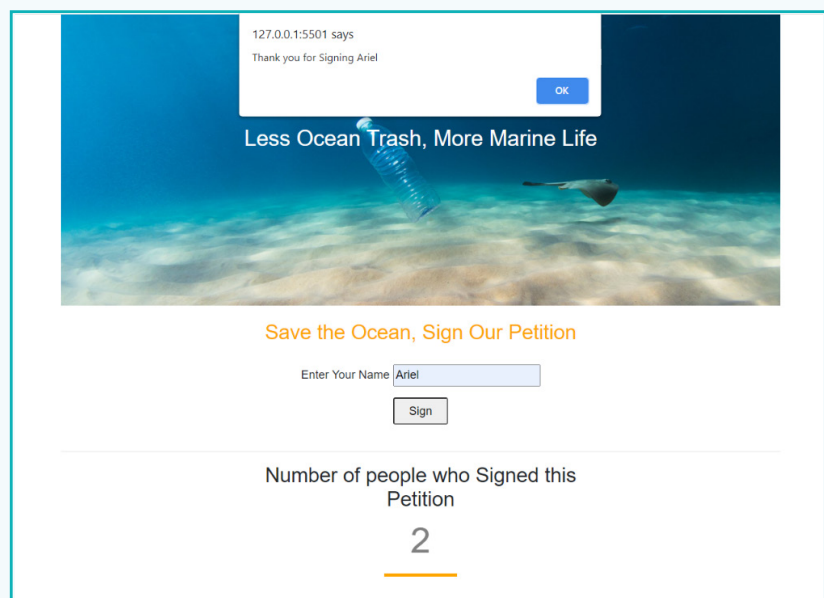
- b. Display the counter value in the `<h2>` heading tag, which has the **ID** of **displayCount**.

```
<div class="petition-counter">
  <h3> Number of people who Signed this Petition</h3>
  <h2 id="displayCount"> 0 </h2>
</div>
```

- c. Access the `<h2>` tag using `'document.getElementById()'` function. To set the value of the counter variable, append the `innerText` property to it. This property is used to add dynamic text on an HTML element.

```
function countPetitions() {
  name = document.getElementById('name').value;
  alert("Thank You for Signing "+name);
  counter = counter + 1;
  document.getElementById('displayCount').innerText = counter;
}
```

Now, run this application on the browser. Enter your name and click on the Sign button. An alert message will pop up. If you click on OK, you should be able to see the value of the count variable as updated to 1. This value will continuously increase with the increase in signatures on the petition.



Exercise

Fill in the blanks

- 1 To declare a variable in JavaScript, the _____ keyword is used.
- 2 _____ refers to a block of code that performs a specific task or operation.
- 3 _____ is a pre-defined function in JavaScript that displays a pop-up box.
- 4 To access an HTML element with a specific ID, we use the _____ function.

JavaScript is the most commonly used programming language to add interactivity to a website. It enables us to add interactive elements to a website, which enhances a user's entire experience. In this lesson, we learnt to create a dynamic element on a webpage in the form of a petition.

We add forms, comment sections, and animations to a website to increase its interactivity.



Tech Flash

- **Dynamic Functionality** : A concept that involves developing web applications with interactive features like using Forms to collect user data.